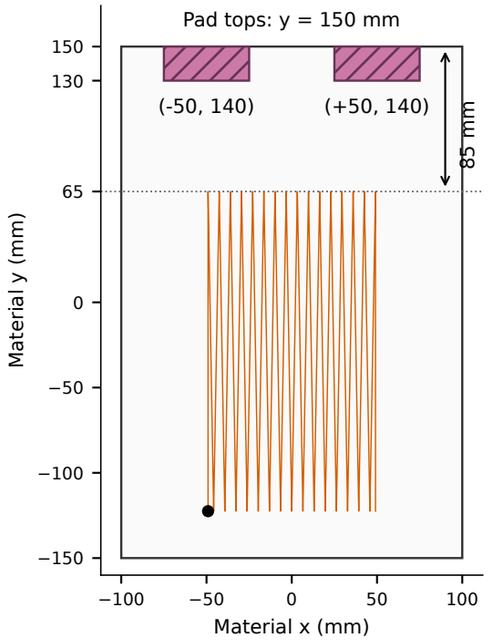
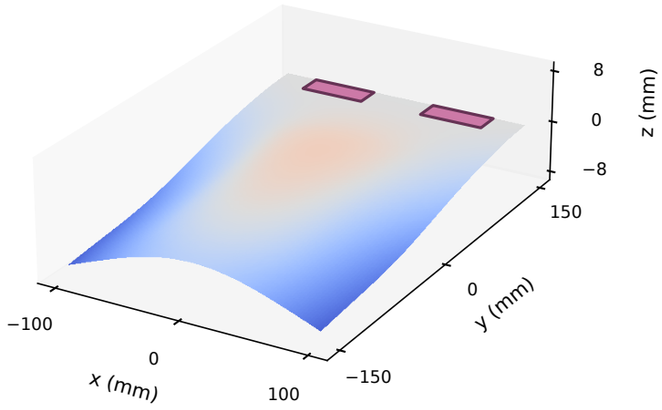
(a) Run 9: toolpath + BC



(b) Clamped equilibrium

33 repeats; both pads fixed xyz

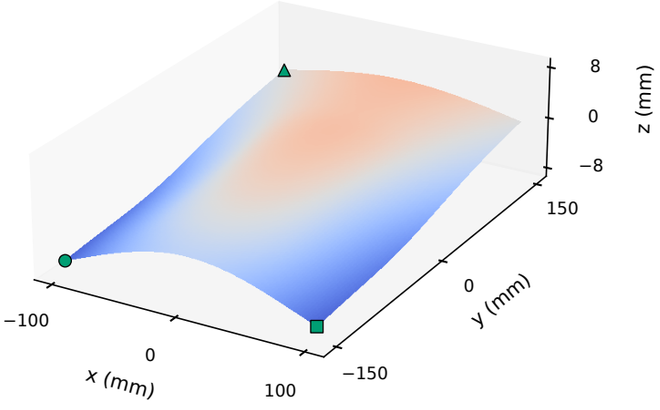


Height ×5; z labels are actual mm

(c) Released: pinned

A: xyz B: yz C: z

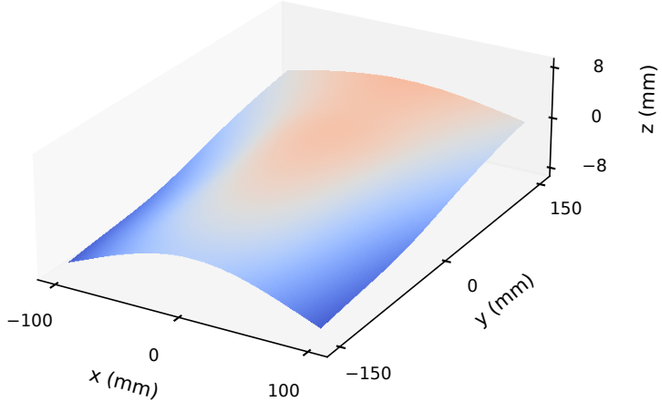
Pads removed; six coordinates retained



Height ×5; z labels are actual mm

(d) Released: free

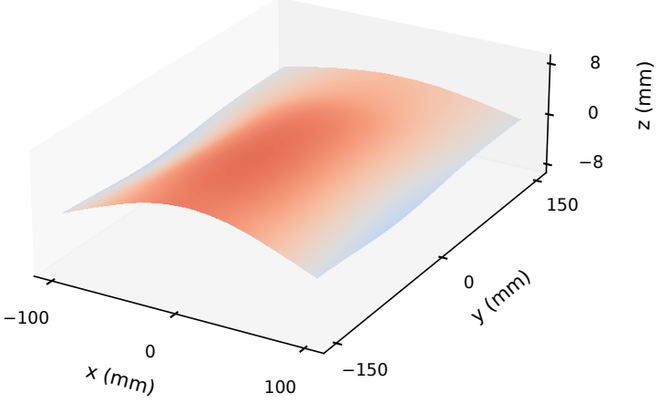
Pads removed; no fixed vertices



Height ×5; z labels are actual mm

(e) Never-clamped per-cycle

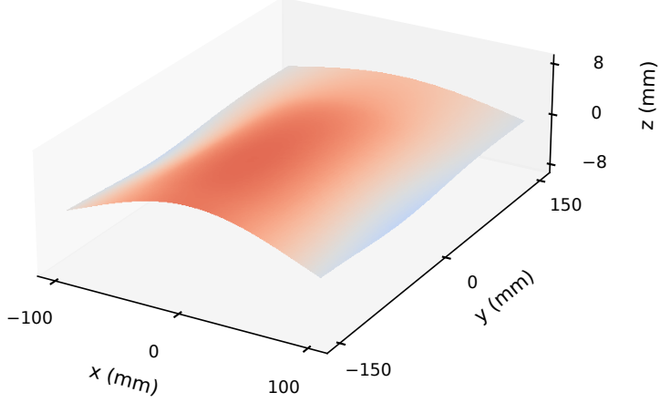
No pads or persistent pins during forming



Height ×5; z labels are actual mm

(f) Never-clamped final-only

No pads or persistent pins during forming



Height ×5; z labels are actual mm

